



Universität
Zürich^{UZH}

Modelling Reaction Times and Accuracies Simultaneously

Gidon Frischkorn, Ruhi Bhanap and Isabel Courage

Satellite Event: Analysis at the Level of Cognitive Processes

ESCoP 2025, Sheffield

Workshop: Overview

Monday

Tuesday

Morning

Intro
Why
hierarchical
models?

Signal-
Detection
Models
(brms)

Diffusion
Decision
Model
(bmm)

Practical
Exercises

Materials
are available
via GitHub

30 Min. Coffee
Breaks
10:30 & 15:00
60 Min. Lunch
Break
12:30

Afternoon

Memory
Measureme
nt Model
(bmm)

Practical
Exercises

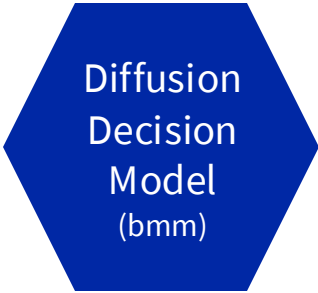
Signal
Discrimination
Model

Reporting
Results from
Cognitive
Modelling
Analyses

Ask
questions
whenever
something is
not clear



Overview



Diffusion
Decision
Model
(bmm)

- Introducing the Drift Diffusion Model
- Simplified Estimation Techniques for the Drift Diffusion Model (the EZ Diffusion Model, bmm)
- Estimating the full Drift Diffusion Model with bias for Response Alternatives

So Far.....

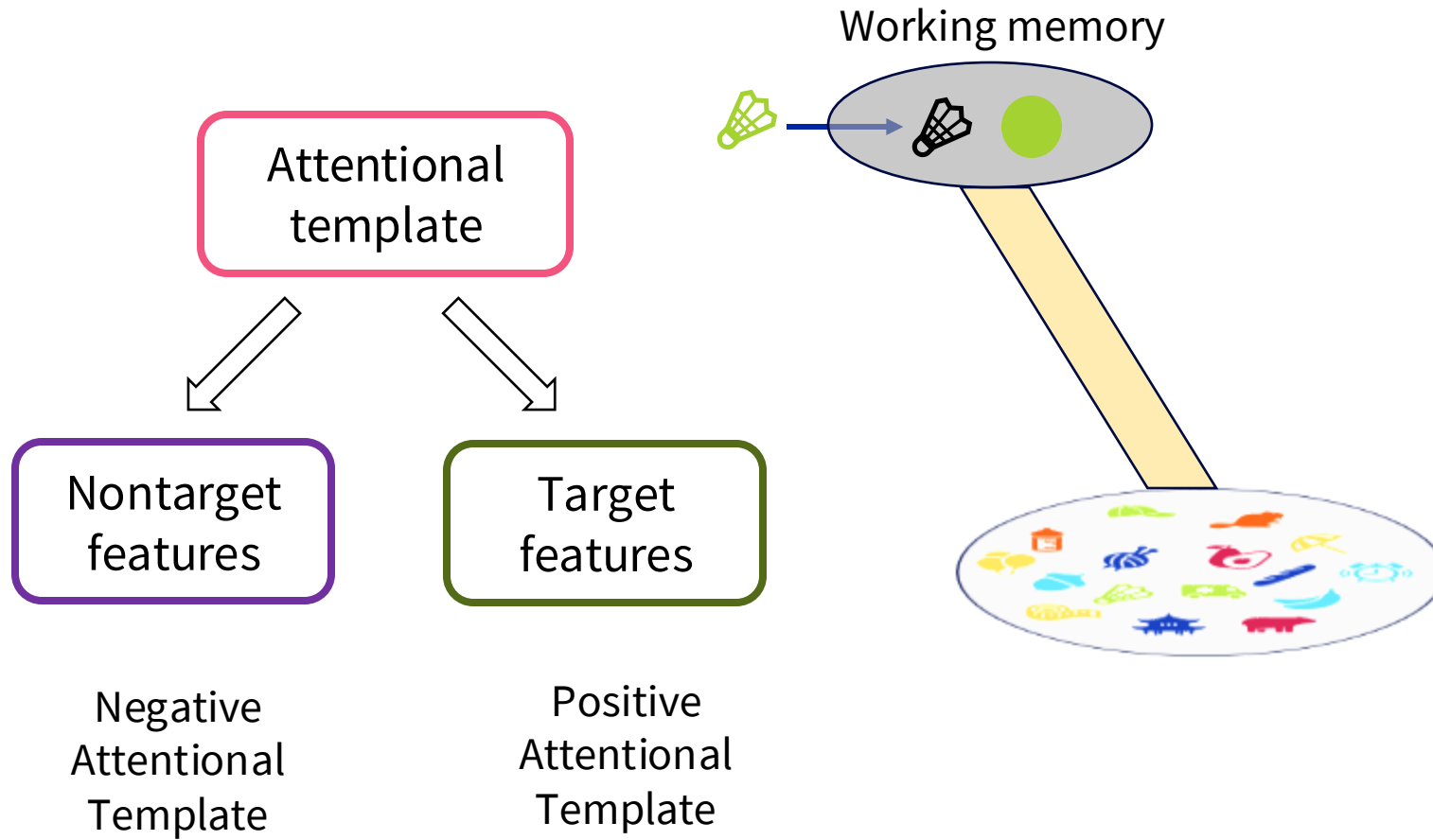
- ✓ Beyond Accuracy Rates
- ✓ Types of Errors Matter

+

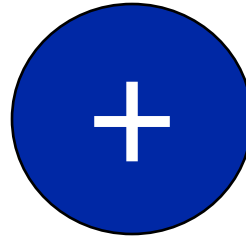
Reaction Times

Introduction: The Visual Search Task

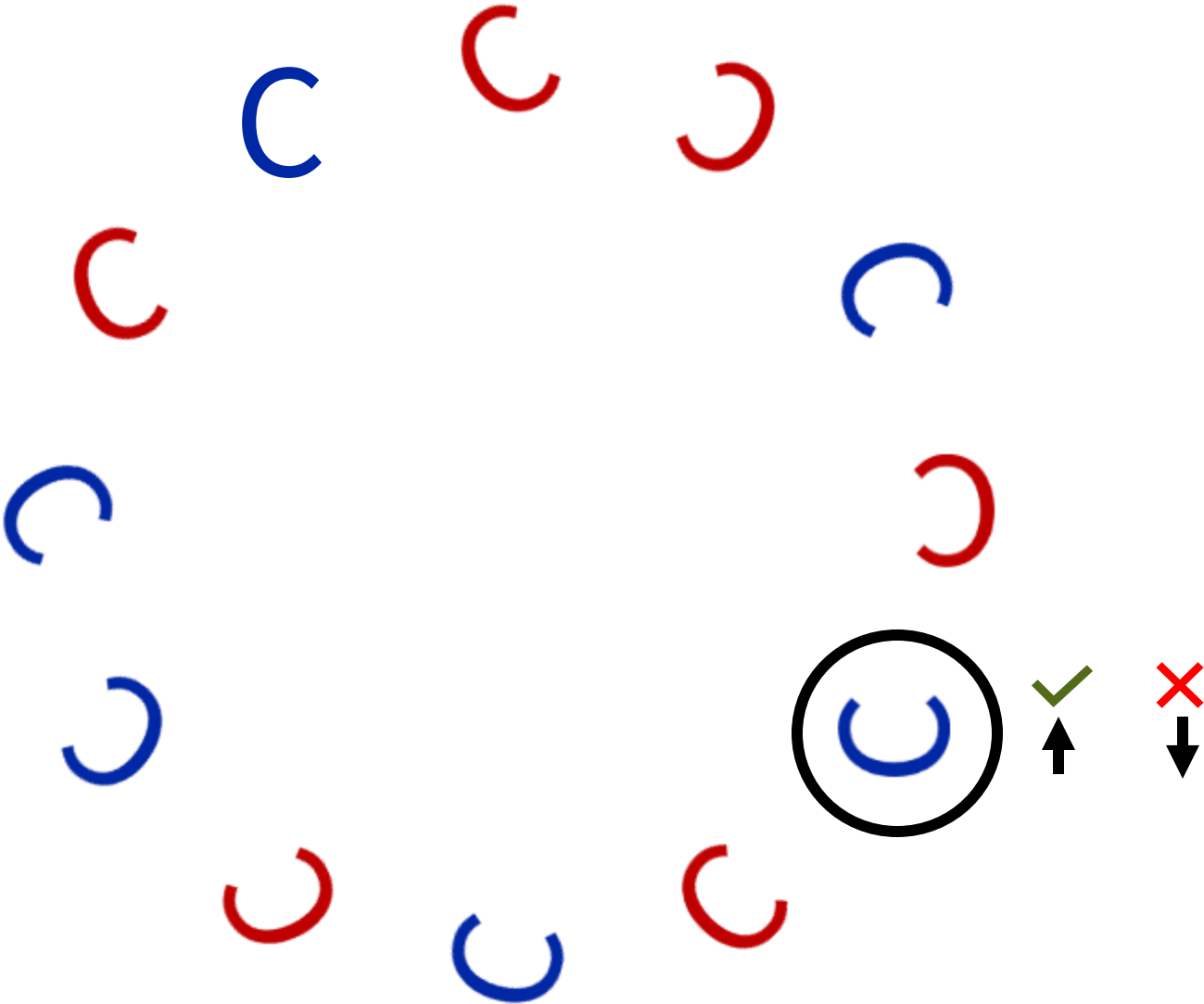
Diffusion
Decision
Model
(bmm)



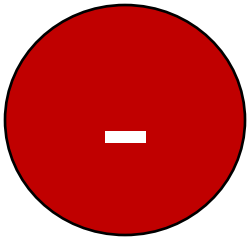
Experimental Task



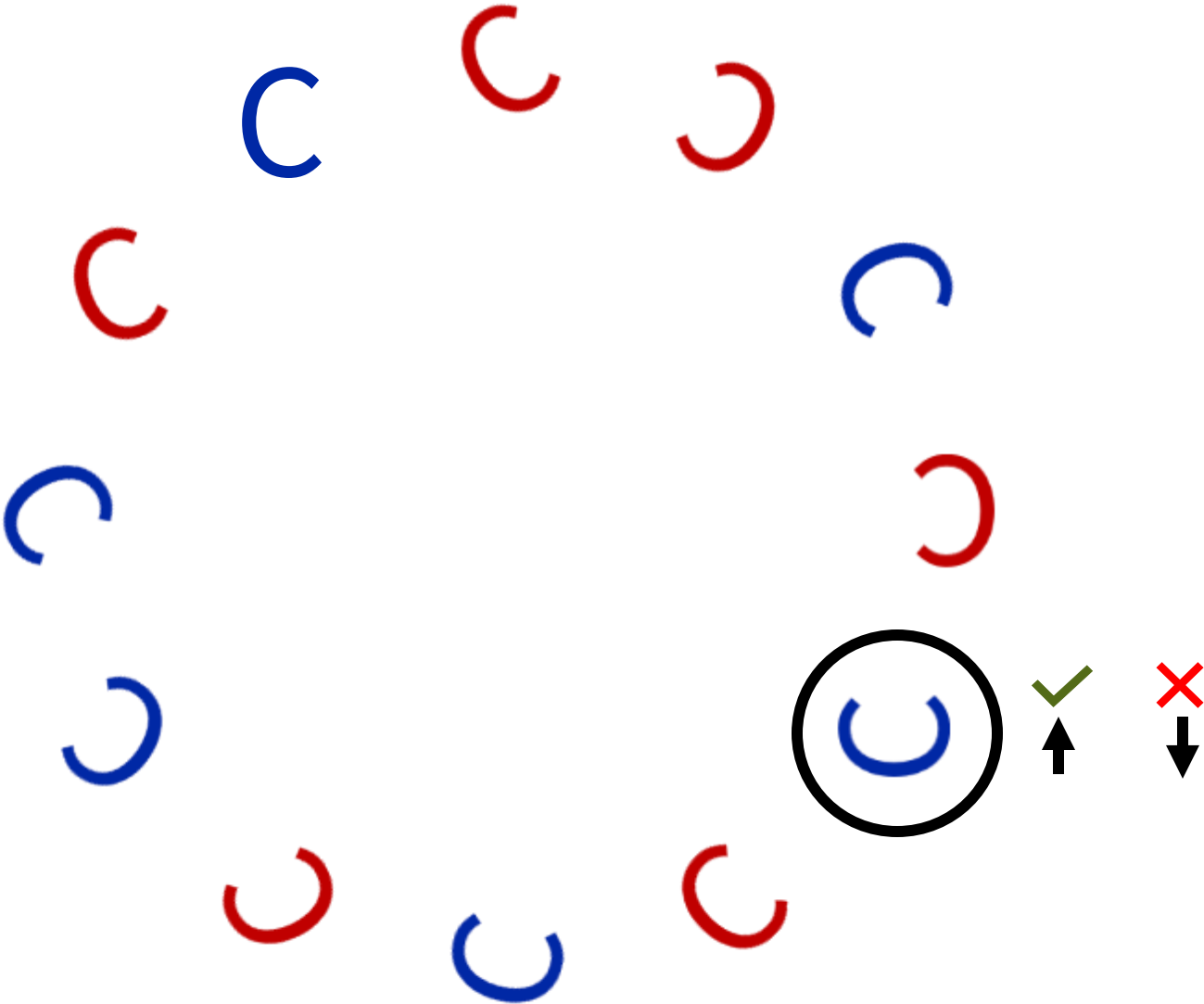
Experimental Task



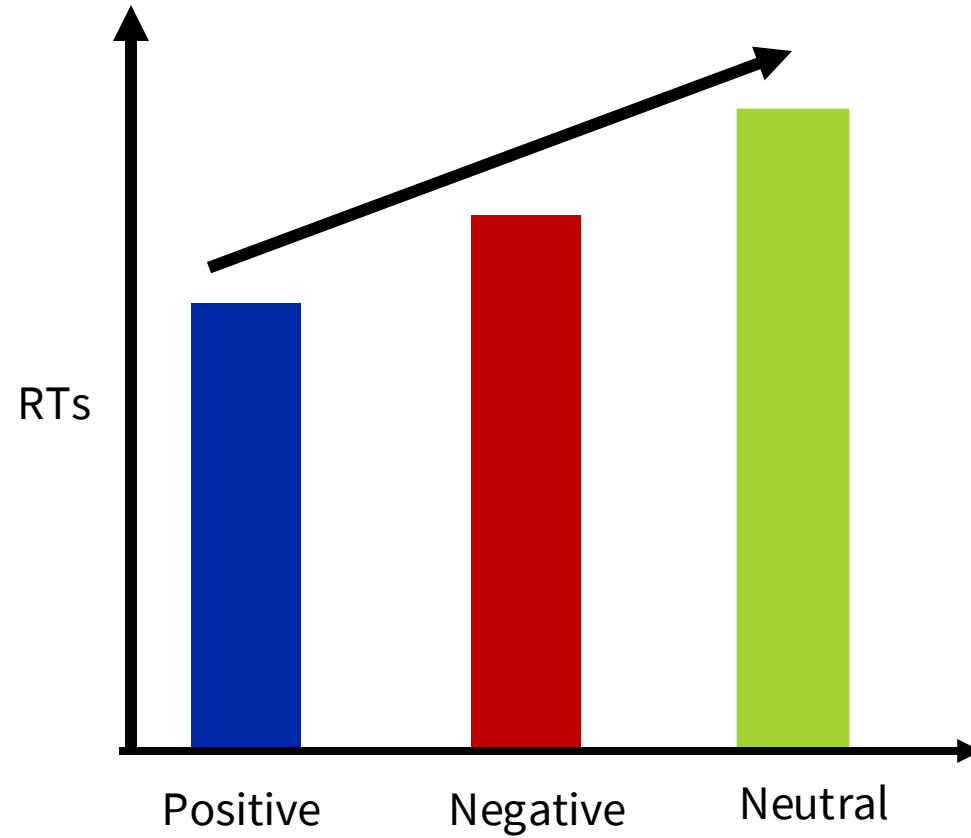
Experimental Task



Experimental Task



Typical Results



Reaction Times can inform about the cost induced by having an informative vs. non informative template.

Why model both RTs and accuracies?

Positive Template

1000 ms

Negative Template

1200 ms

Neutral Template

1500 ms

Why model both RTs and accuracies?

Positive Template

1200 ms

90%

Negative Template

1200 ms

70%

Neutral Template

1300 ms

50%

What do the reaction times tell us?

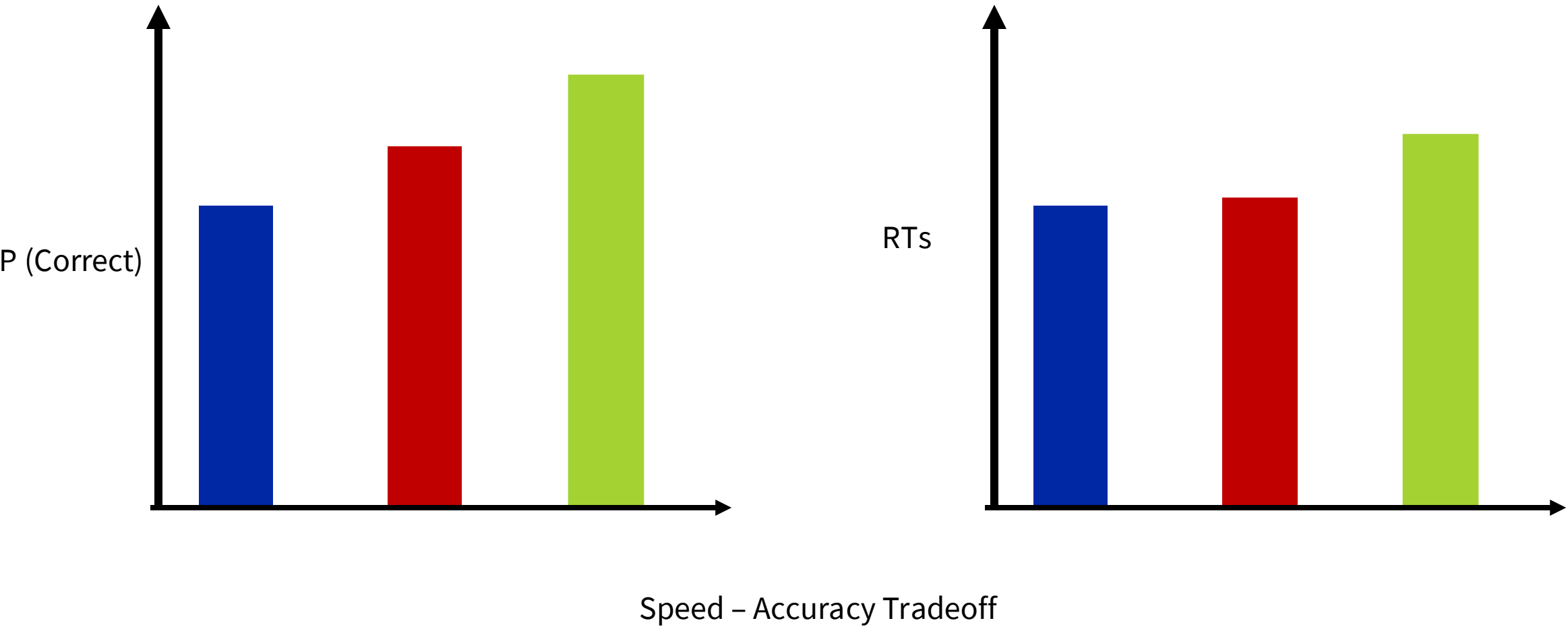
Positive and Negative Template guide attention similarly!

What do the accuracies tell us?

Positive Templates are more effective in guiding attention!

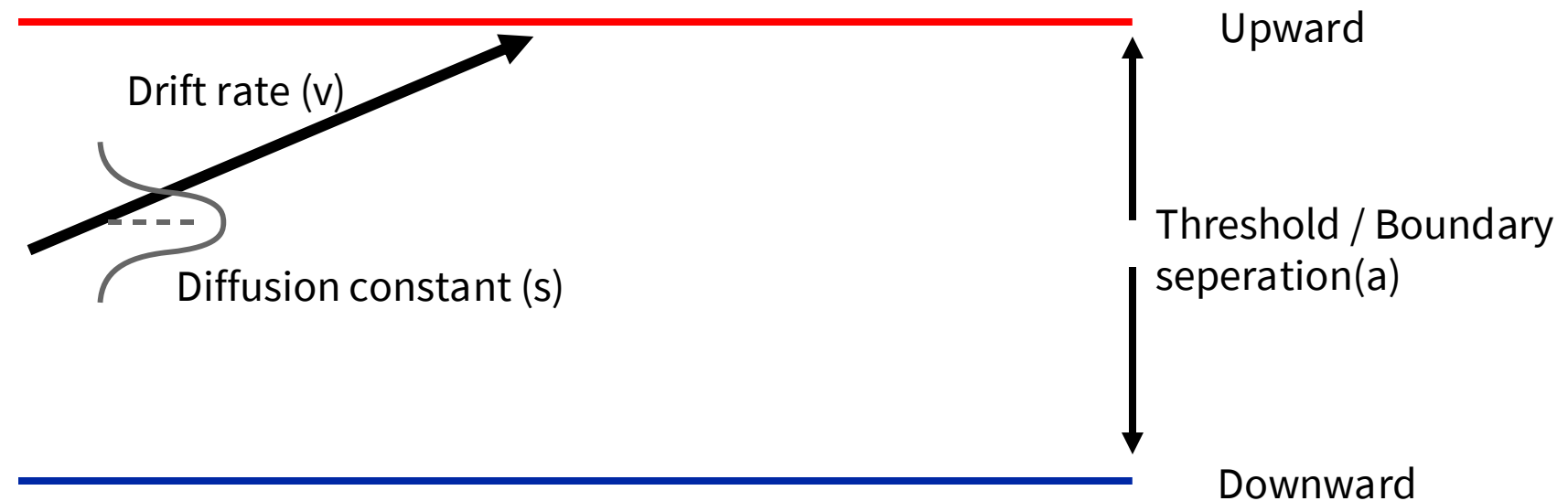
Which conclusion will you derive from these results?

Why model RT and Accuracies together?



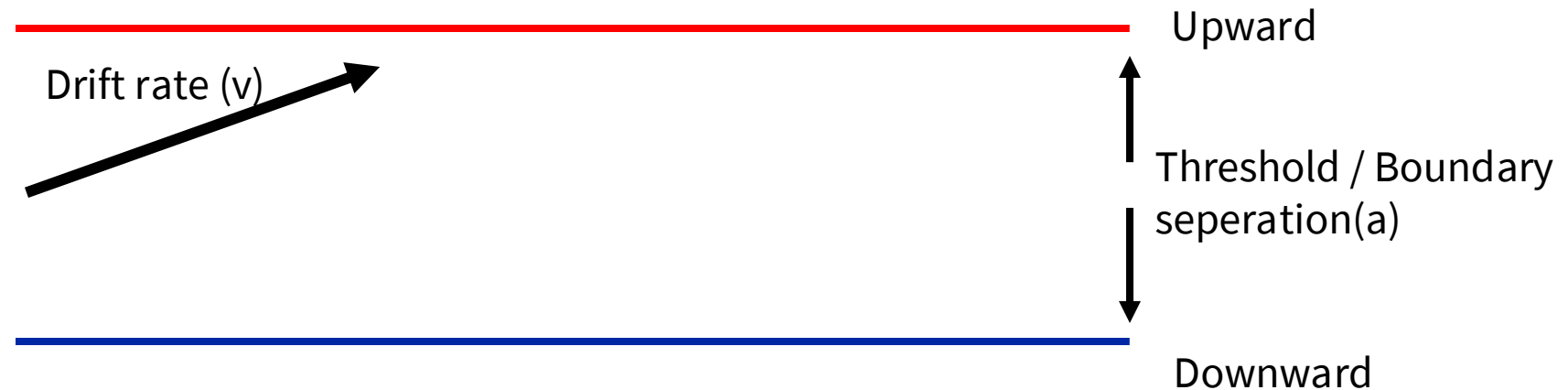
What is a drift diffusion model?

- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision



What is a drift diffusion model?

- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision

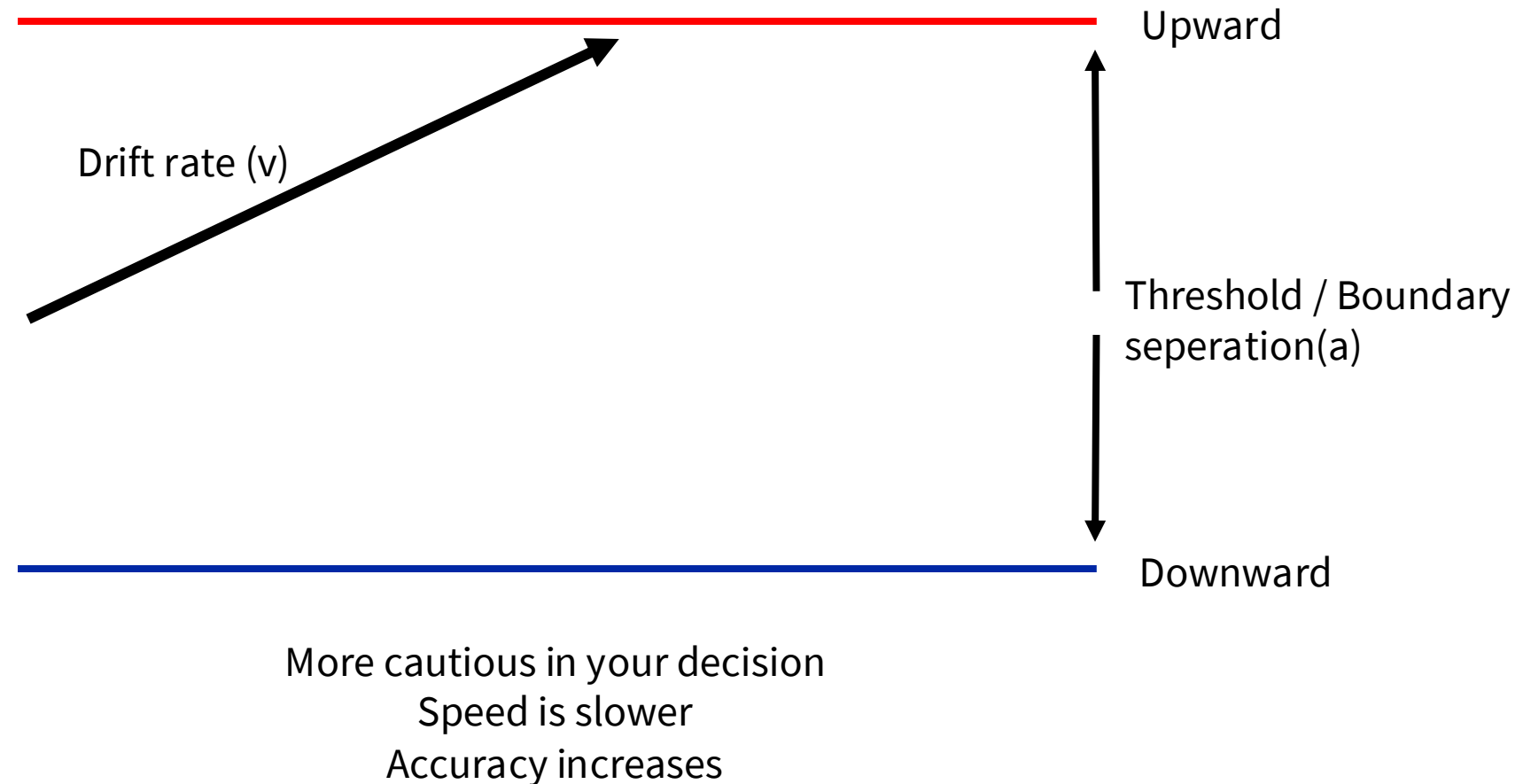


Risky in your decision
Speed is faster
Accuracy reduces

What is a drift diffusion model?

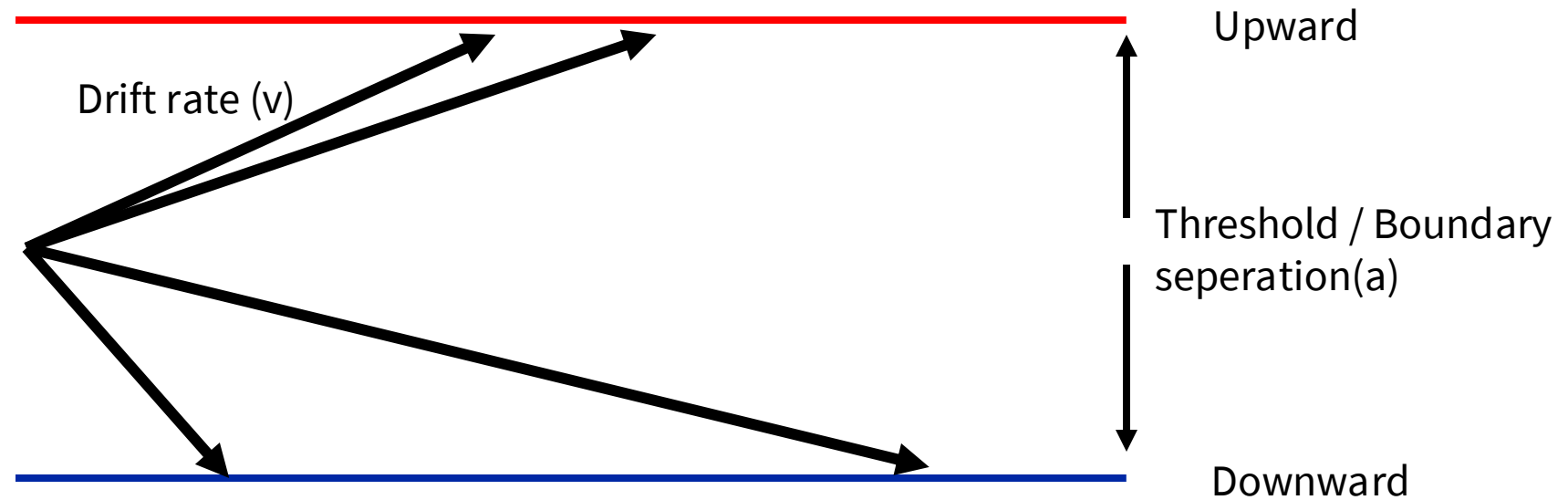
Diffusion
Decision
Model
(bmm)

- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision



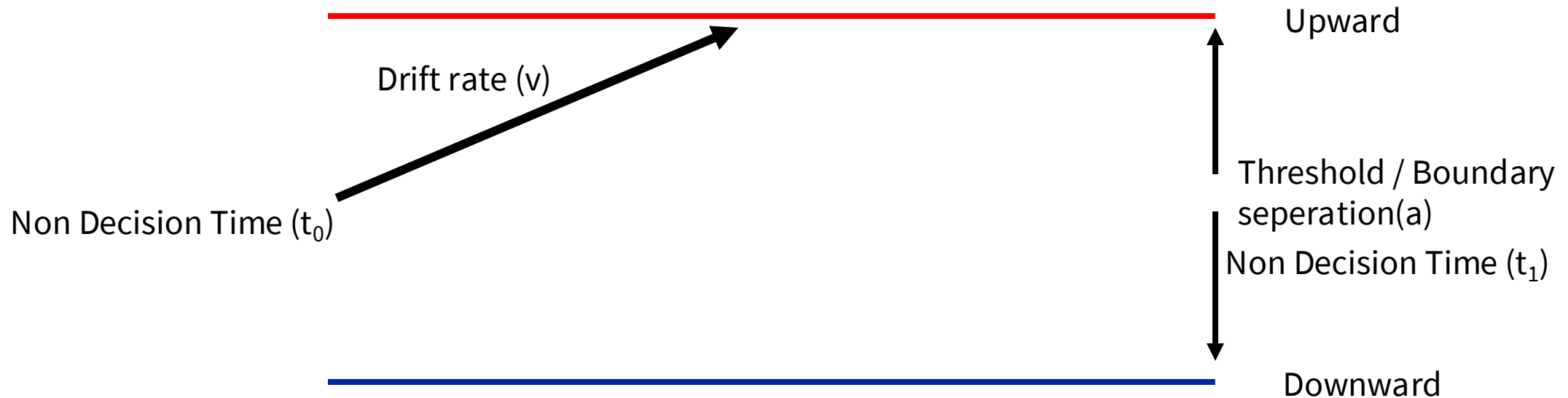
What is a drift diffusion model?

- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision



What is a drift diffusion model?

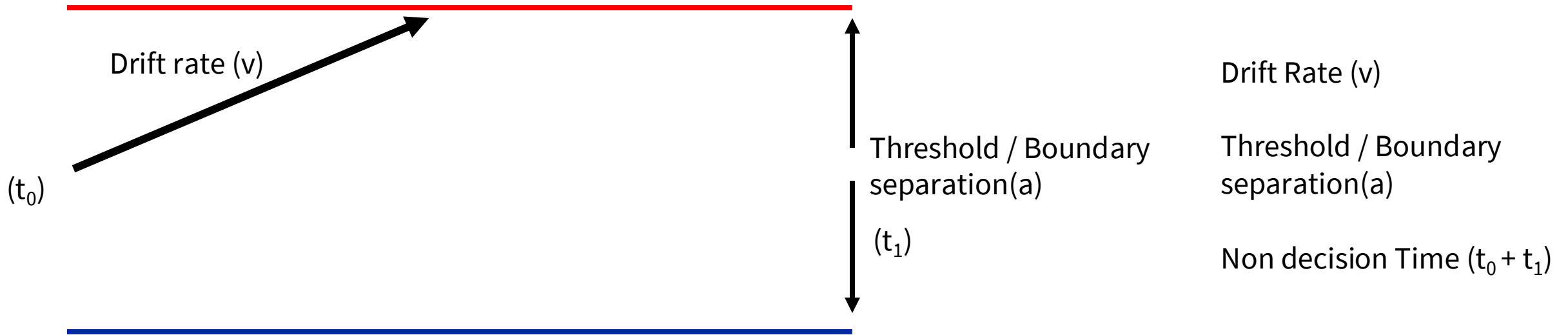
- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision



What is a drift diffusion model?

- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision

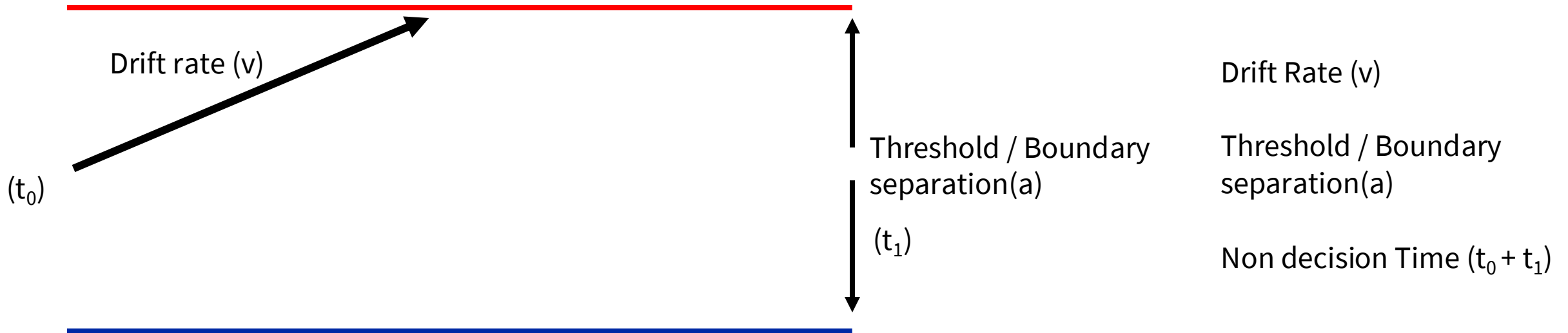
Diffusion
Decision
Model
(bmm)



What is a drift diffusion model?

- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision

Diffusion
Decision
Model
(bmm)

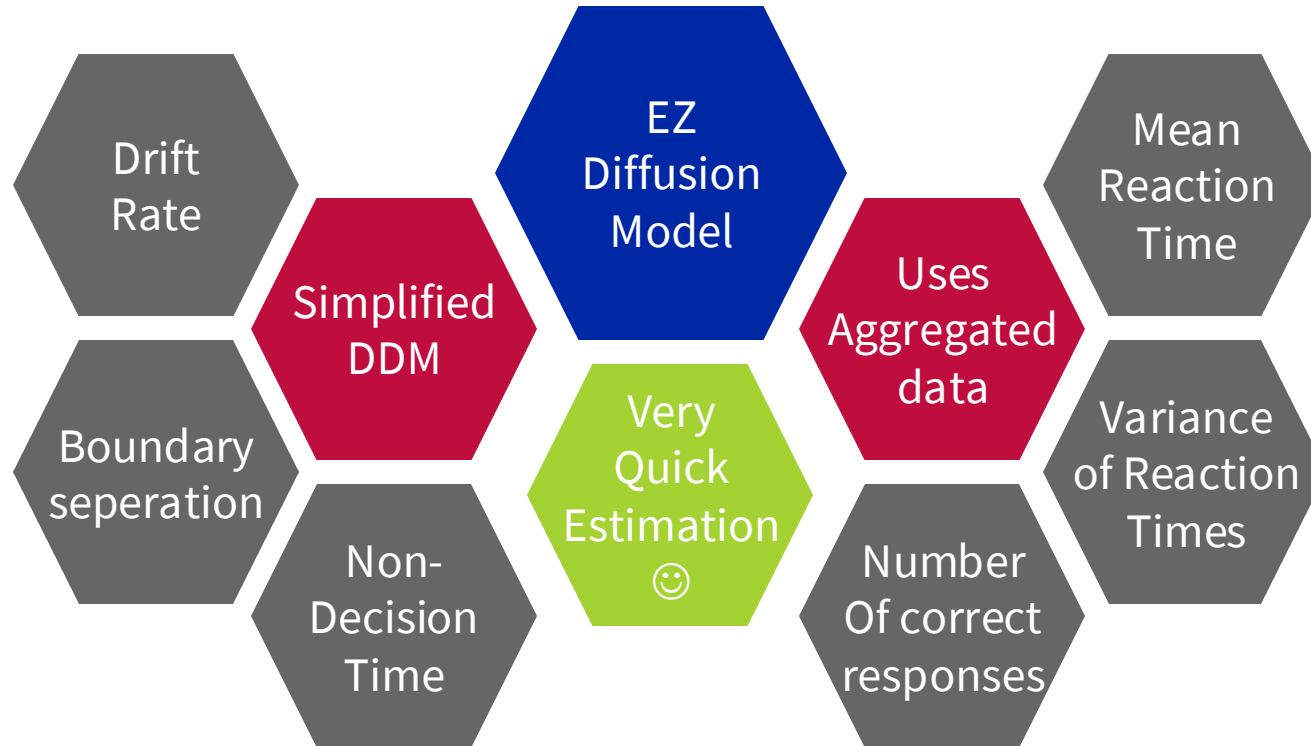
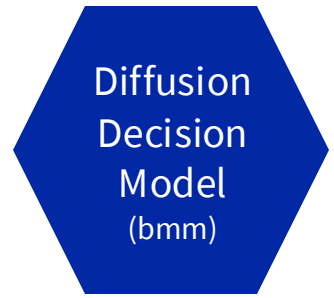




What are your questions so far?

How to implement Diffusion models in bmm?

EZ Diffusion Model



How to implement Diffusion models in bmm?

EZ Diffusion Model – Model Object

- Again, we set up a *bmmmodel* object for the EZ-Diffusion model using the `ezdm()` function:

```
ezdm_model = ezdm(
```

```
    mean_rt,
```

```
    var_rt ,
```

```
    n_correct,
```

```
    n_trials
```

```
)
```

Mean Reaction Times for each participant at each level of condition

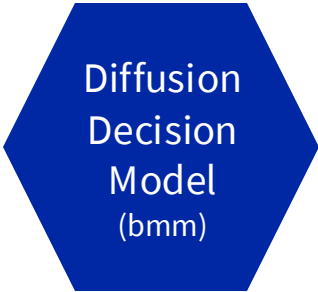
Variance of Reaction Times for each participant at each level of condition

Total Number of Correct Responses

Number of trials in each cell of the experimental design

How to implement Diffusion models in bmm?

EZ Diffusion Model – bmmformula

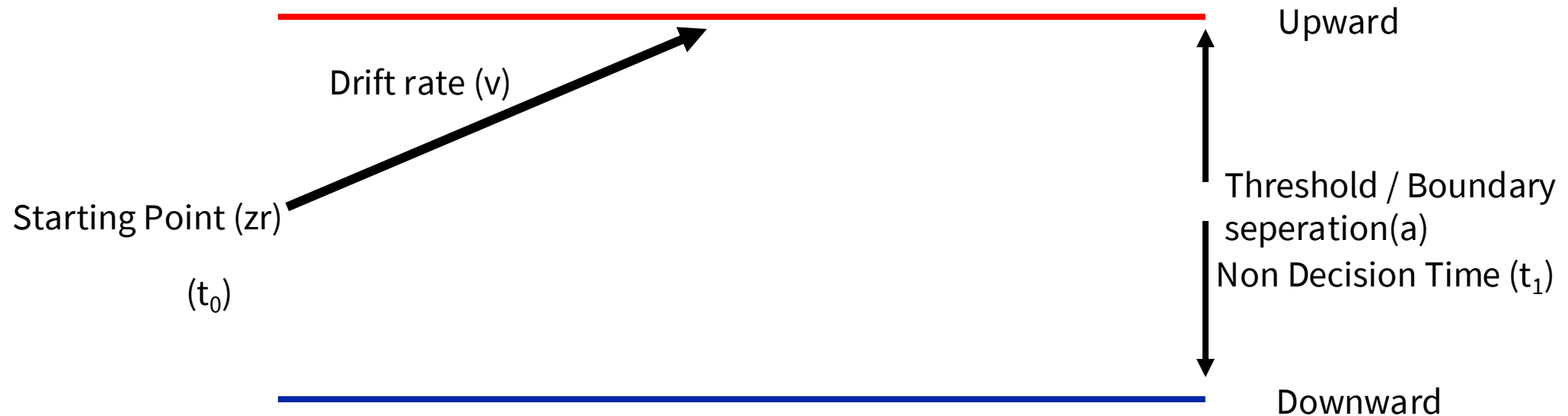


```
ezdm_formula <- bmf(  
  drift ~ 1 + condition + (1 + condition | participant),  
  
  bound ~ 1 + (1 | participant),  
  
  ndt ~ 1 + condition + (1 + condition | participant)  
)
```

	Drift Rate	Bound	Non Decision Time
Template Condition	Positive > Negative	Positive = Negative	Positive > Negative

What is a drift diffusion model?

- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision



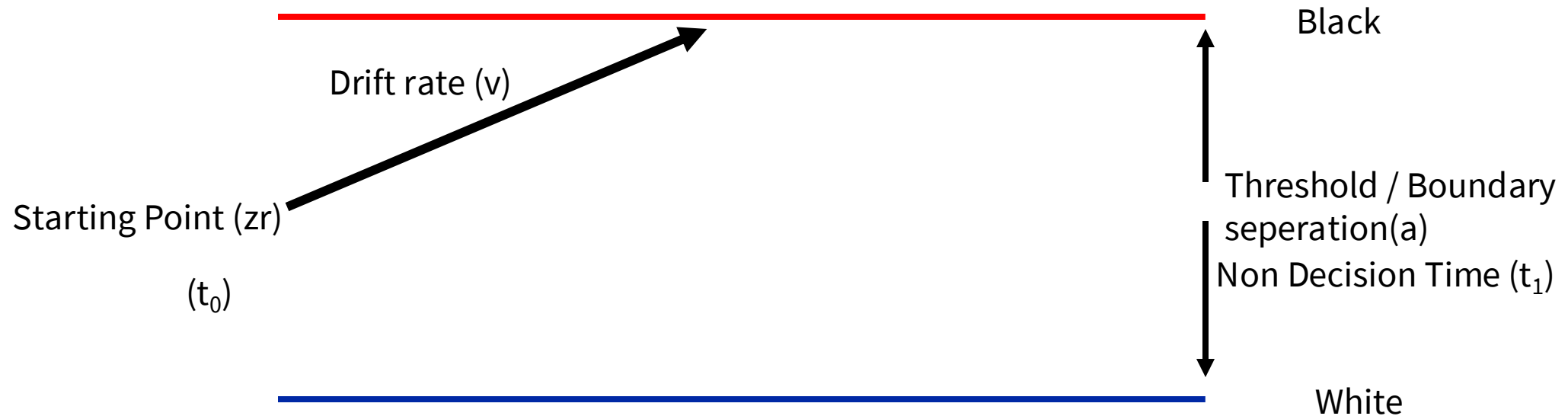
Perceptual Decision Making Task

Diffusion
Decision
Model
(bmm)



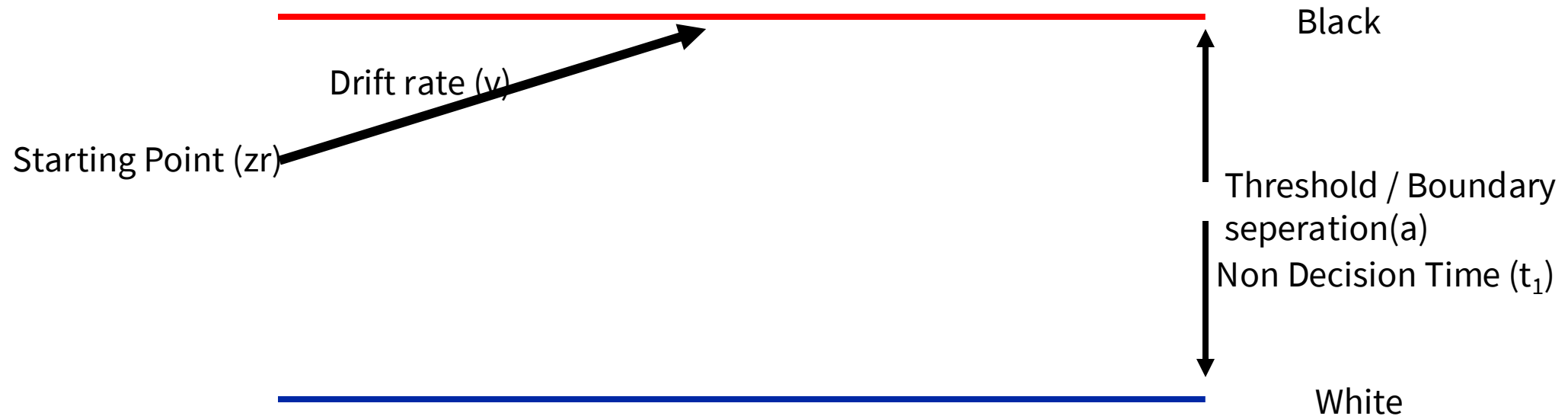
What is a drift diffusion model?

- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision



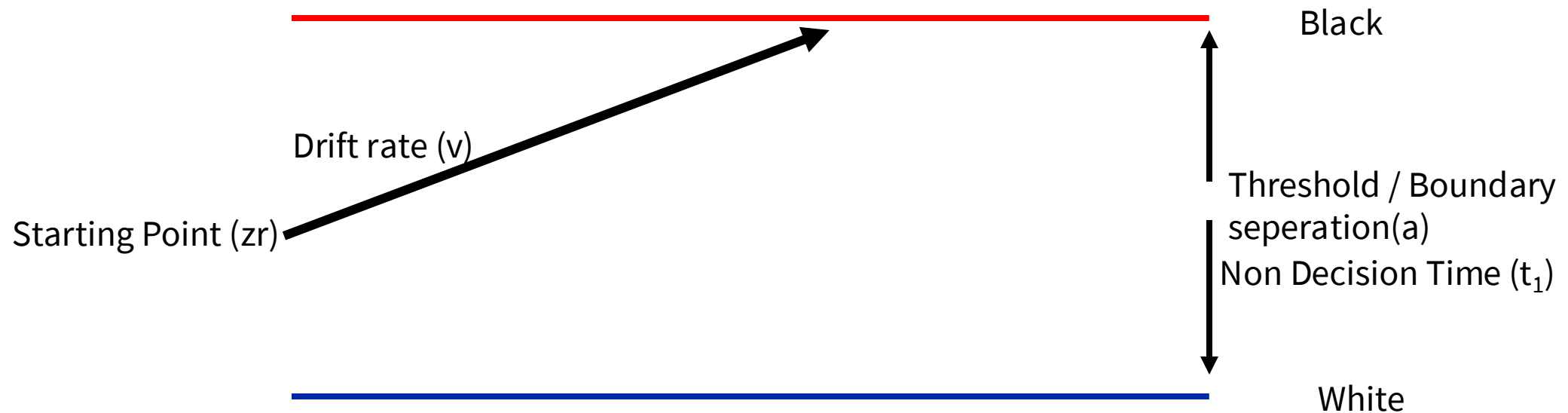
What is a drift diffusion model?

- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision



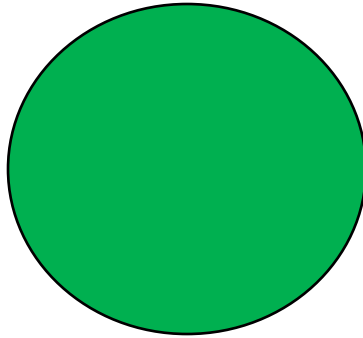
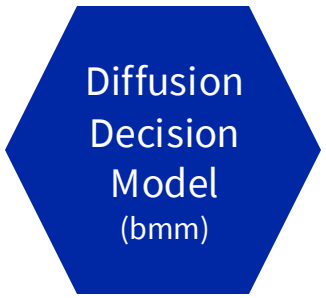
What is a drift diffusion model?

- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision



Example of bias – Attentional ddm

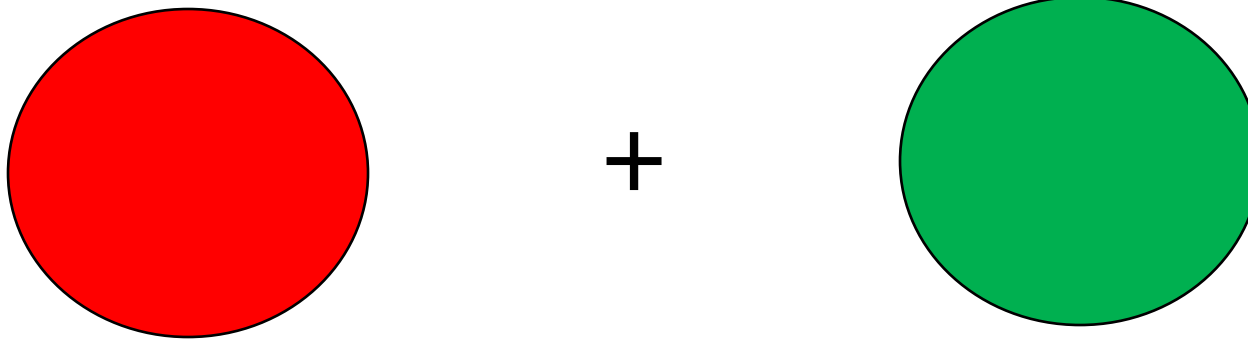
- Leftward bias in eye movements



Example of bias – Attentional ddm

- Leftward bias in eye movements

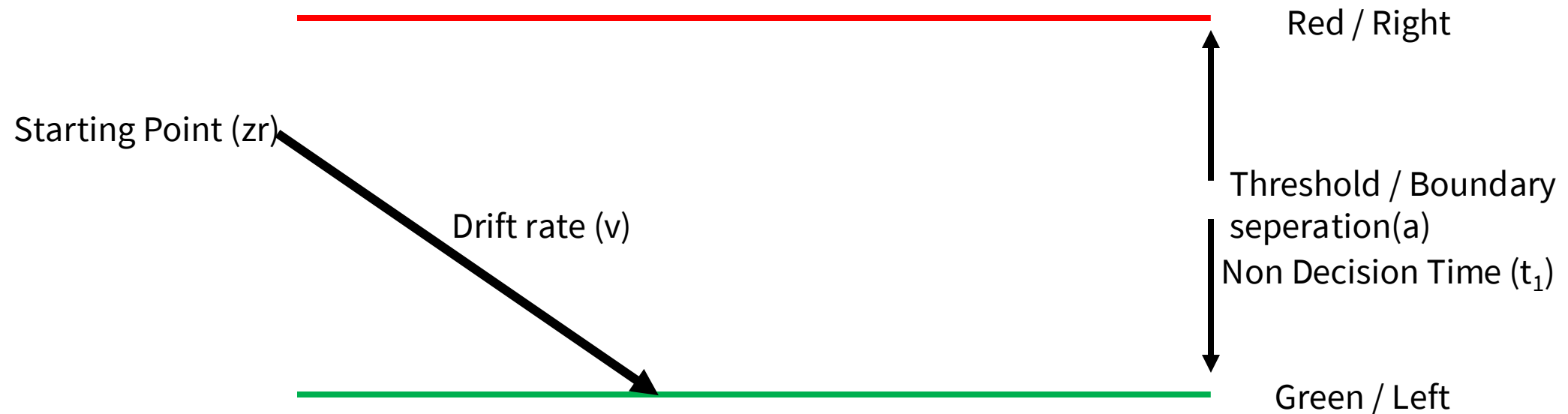
Diffusion
Decision
Model
(bmm)



Which one did you memorize?

What is a drift diffusion model?

- Falls under the widely known category of evidence accumulation models.
- Cognitive System sampling information from the visual space to make a decision

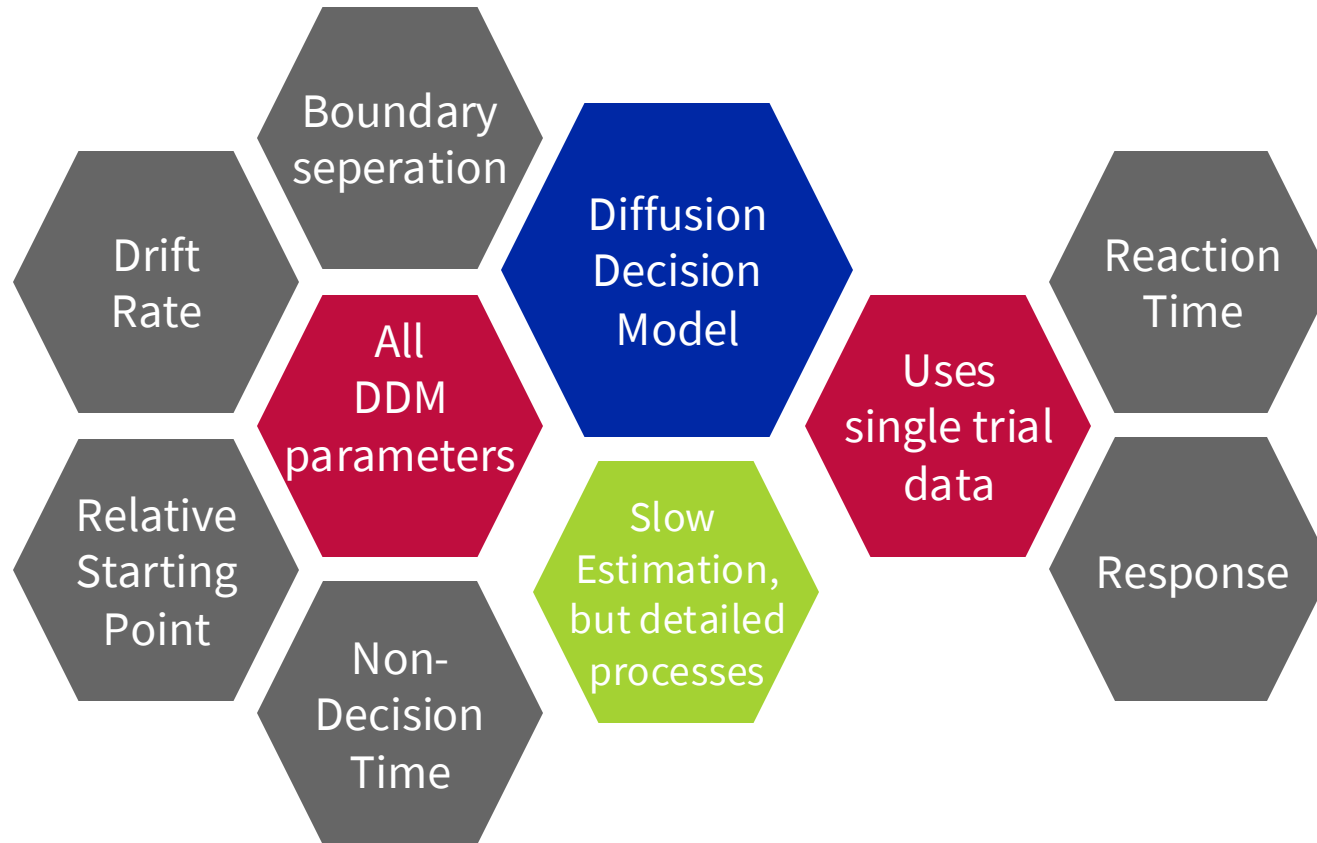
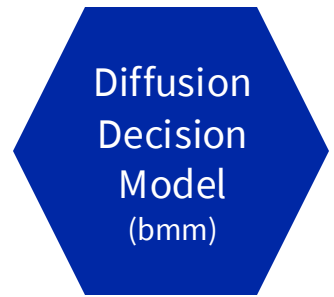




What are your questions so far?

How to implement Diffusion models in bmm?

Full Diffusion Decision Model (DDM)



How to implement Diffusion models in bmm?

Full Diffusion Decision Model (DDM) – Model Object

- Again, we set up a *bmmmodel* object for the Diffusion Decision model using the `ddm()` function:

```
ddm (
```

```
  rt,
```

```
  response,
```

```
  version
```

```
)
```

Name of the **reaction time variable** coding reaction time in seconds in the data.

Name of the **response variable** coding the response numerically

→ 0 = lower response / incorrect, 1 = upper response / correct

version = three_par (drift, bound, ndt)

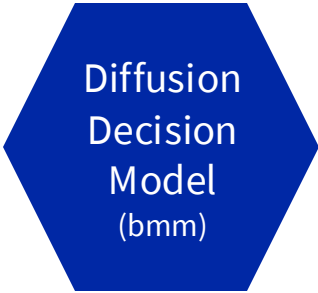
four_par (drift, bound, ndt, zr)

seven_par (drift, bound, ndt, zr, sdrift, sndt, szr)*

* Trial by Trial Variability in any of these parameters

How to implement Diffusion models in bmm?

Full Diffusion Decision Model (DDM) – bmmformula



```
ezdm_formula <- bmf(  
  
  drift ~ 0 + target:condition + (0 + target:condition | participant),  
  
  bound ~ 1 + (1 | participant)  
  
  ndt ~ 0 + condition + (0 + condition | participant)  
  
  zr ~ 1 + (1 | participant)  
  
)
```

	Drift Rate	Bound	Non Decision Time	Bias
Difficulty Condition	High < low	None	High < low	No
Target Response	Black = -White	None	None	No

How do we assess the ddm in bmm?

- `summary()`
show the model output
- `plot()`
plots posterior distributions for every model estimate and the posterior samples per chain
→ check model convergence
- `hypothesis()`
performs hypothesis test → $\text{Evid.Ratio} = \text{BF}_{01}$



What are your questions so far?

References

- Bürkner, P. C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of statistical software*, 80, 1-28.
- Carlisle, N. B. (2023). Negative and positive templates: Two forms of cued attentional control. *Attention, Perception, & Psychophysics*, 85(3), 585-595.
- Frischkorn, G. T., & Popov, V. (2025). A tutorial for estimating Bayesian hierarchical mixture models for visual working memory tasks: Introducing the Bayesian Measurement Modeling (bmm) package for R. *Behavior Research Methods*, 57(5), 144
- Tavares, G., Perona, P., & Rangel, A. (2017). The attentional drift diffusion model of simple perceptual decision-making. *Frontiers in neuroscience*, 11, 468.